

ZEYU (STEVEN) HE

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RESEARCH INTERESTS

I build the interfaces, models, and pipelines that capture **high-fidelity human judgment** and turn it into training and evaluation signal for AI systems. My work spans **human data and annotation pipelines**, **LLM evaluation and LLM-as-judge validity**, **individual-level preference simulation**, and **human oversight of agentic systems**, at the intersection of **NLP** and **HCI**.

EDUCATION

Pennsylvania State University, University Park

Aug 2022 - Jan 2027 (expected)

Ph.D. Candidate, Informatics (College of Information Sciences and Technology)

Advisor: Dr. Ting-Hao ‘Kenneth’ Huang

University of California, Irvine

Sep 2016 - June 2020

B.S. Computer Science and Engineering; B.S. Civil Engineering

GPA: 3.74/4.0

PUBLICATIONS

Peer-Reviewed Conference and Journal Papers

1. **Zeyu He**, Xuan Qi, Subramanian Chidambaram, Zhichao Xu, Vinayak Arannil, Lydia Chilton, Alex C. Williams. *PersonaJudge: Simulating Individual Human Preference Judgments with Evaluator-Specific Demonstration Data*. **SIGDIAL 2026**. (Amazon internship)
2. **Zeyu He**, Hannah Kim, Dan Zhang, Estevam Hruschka. *How To Steer Your Multi-Agent System: Human-LLM Collaborative Planning*. **ACM CAIS 2026**. **Best Artifact Award**. (Megagon Labs internship)
3. **Zeyu He**, Saniya Naphade, Ting-Hao K. Huang. *Prompting in the Dark: Assessing Human Performance in Prompt Engineering for Data Labeling When Gold Labels Are Absent*. **ACM CHI 2025**.
4. **Zeyu He**, Chieh-Yang Huang, Chien-Kuang (Cornelia) Ding, Shaurya Rohatgi, Ting-Hao ‘Kenneth’ Huang. *If in a Crowdsourced Data Annotation Pipeline, a GPT-4*. **ACM CHI 2024**.
5. Wenqian Ye, Luyang Jiang, Eric Xie, . . . , **Zeyu He**, . . . , Henry Kautz, Aidong Zhang. *The Clever Hans Mirage: A Comprehensive Survey on Spurious Correlations in Machine Learning*. **TMLR 2026**.
6. Jakob Hederich, Shreya Ghosh, **Zeyu He**, Prasenjit Mitra. *Understanding the Night-Sky? Developing AI-Enabled System for Exploring Night-Light Usage Patterns*. **IJCAI 2023**.
7. Lyuyang Hu, Omkar Pathak, **Zeyu He**, Hunkyu Lee, Mina Bedwany, Jace Mica, Peter J. Burke. *“CloudStation.” A Cloud-Based Ground Control Station for Drones*. **IEEE Journal on Miniaturization for Air and Space Systems**, 2(1):36–42, 2020.

Manuscripts Under Review

8. **Zeyu He**, Zhuqian Zhou, Kirk Vanacore, Rene F. Kizilcec, Ting-Hao ‘Kenneth’ Huang. *Experts Rise Where LLMs Disagree: Using Cross-Model Disagreement to Target Expert Effort in LLM Codebook Revision for Large-Scale Annotation*. Under review, **ACM CHI 2027**.

Workshop Papers, Posters, and Doctoral Consortium

9. **Zeyu He**. *Human Involvements in Data Annotation in the Era of Large Language Models*. **HCOMP 2024** Doctoral Consortium.
10. Ho Yin Ng, **Zeyu He**, Ting-Hao K. Huang. *What Color Scheme is More Effective in Assisting Readers to Locate Information in a Color-Coded Article?* **IEEE VIS 2024**, Short Paper.
11. Shih-Hong Huang, Ya-Fang Lin, **Zeyu He**, Chieh-Yang Huang, Ting-Hao K. Huang. *How Does Conversation Length Impact User's Satisfaction? A Case Study of Length-Controlled Conversations with LLM-Powered Chatbots*. **ACM CHI 2024**, Late-Breaking Work.

RESEARCH EXPERIENCE

Ph.D. Residency (Research)

Google [X], the Moonshot Factory
Host: Dr. Bertrand Delorme

July 2026 - Dec 2026
Mountain View, CA

- Research on LLM-based idea generation and evaluation for early-stage project assessment.

Visiting Researcher

Cornell University, Department of Information Science

Collaborators: Dr. Rene Kizilcec, Dr. Kirk Vanacore (Cornell Future of Learning Lab)

Jan 2026 - July 2026
Ithaca, NY

- Developed methods that use expert disagreement to drive codebook and annotation-rubric refinement for tutor-intention coding; basis of a first-author CHI 2027 submission.
- Grounded the methods in expert educational-transcript coding practice and learning-science measurement standards through collaboration with the Future of Learning Lab.

Research Scientist Intern

Megagon Labs

Hosts: Dr. Hannah Kim, Dr. Dan Zhang

Aug 2025 - Nov 2025
Mountain View, CA

- Formalized a design space for human-LLM collaborative planning in orchestrated multi-agent systems along three axes (mode, scope, level), making competing co-planning strategies directly comparable.
- Built AMBIPOM, a prototype supporting process-level supervision of multi-agent plans through both semantic and structural interaction; ran a user study plus a controlled benchmark of LLM plan revision.
- Published at ACM CAIS 2026 and received the **Best Artifact Award**; released [code](#) and [data](#) on GitHub.

Applied Scientist Intern

Amazon (Bedrock / AWS AI Fundamental Research)

Host: Dr. Alex C. Williams Mentor: Dr. Xuan Qi

May 2025 - Aug 2025
Santa Clara, CA

- Designed and shipped a full-stack annotation platform capturing three complementary evaluator signals — categorical judgments, interface telemetry, and retrospective reasoning — from 32 trained annotators across 4,200 preference judgments.
- Built PersonaJudge, an evaluator-specific simulation framework; a $4 \times 4 \times 4$ factorial study showed up to 9.9-point accuracy gains over a base LLM judge, with reasoning traces helping and interface telemetry hurting.
- Established that evaluator simulatability is systematic and cross-task stable ($r = 0.728$), predicted by neutral-response usage and divergence from consensus.
- Published at SIGDIAL 2026.

Graduate Research Assistant

Pennsylvania State University, College of IST

Jan 2024 - Present
University Park, PA

- Ran the largest holistic comparison of crowdsourced annotation against GPT-4 to date (415 MTurk workers, 127,080 labels, 2 interfaces $\times$ 3 cleaning strategies $\times$ 8 aggregation algorithms): the best crowd pipeline peaked at 81.5% against GPT-4's 83.6%, while aggregating crowd and GPT-4 labels together reached 87.5% — evidence that human value lies in complementary judgment rather than redundant labeling (CHI 2024).
- Built PromptingSheet, a Google Sheets add-on for iterative LLM prompt-based labeling; a 20-participant study showed only 9/20 users improved accuracy over four-plus iterations and that showing LLM explanations induced overreliance (CHI 2025).

INDUSTRY ENGINEERING EXPERIENCE

Software Development Engineer

Amazon Music

Feb 2022 - Aug 2022

Irvine, CA

- Designed and shipped services for Amazon Music at production scale in an agile team; owned CI/CD, infrastructure-as-code, automated testing, and service monitoring.

Software Engineer

Personable Inc.

Aug 2020 - Feb 2022

Fountain Valley, CA

- Built scalable applications for image processing, data extraction, and data management; led QA for monthly releases.

SELECTED SYSTEMS AND ARTIFACTS

- **AMBIPOM** — Agent-aware mixed-initiative block-level interactive planning for orchestrated multi-agent systems. Public [code](#) and [data](#) (CAIS 2026, **Best Artifact Award**).
- **PersonaJudge** — Multi-facet evaluator simulation framework and annotation platform for capturing judgment, telemetry, and retrospective reasoning (SIGDIAL 2026).
- **PromptingSheet** — Google Sheets add-on for spreadsheet-based prompt engineering and iterative LLM data labeling (CHI 2025).

INVITED TALKS AND PRESENTATIONS

- *What Will Be the Human Role in Data Labeling in This Era of LLMs?* Cornell University, Department of Information Science, Feb 2026.

PROFESSIONAL SERVICE

- **Associate Chair:** ACM CHI Late-Breaking Work 2025.
- **Reviewer:** ACM CHI 2026; ACM UIST 2025; ACM CHI LBW 2024; ACM Learning @ Scale 2024; Pluralistic Alignment Workshop @ NeurIPS 2024; ACM TIST 2024.
- **Student Volunteer:** SIGDIAL 2026.

TEACHING EXPERIENCE

Teaching Assistant, Pennsylvania State University

Aug 2022 - Dec 2023

- IST 554: Network Management; DS 220: Data Management for Data Science; DS 300: Privacy in Data Science.

Academic Tutor (Remote), Gravitee Tutor Alliance

Sep 2020 - Dec 2021

- Tutored data structures, AI, and machine learning; adapted instruction to individual learning styles.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Kotlin, SQL, JavaScript/TypeScript, R, MATLAB
ML & LLMs: PyTorch, Hugging Face Transformers, scikit-learn, DSPy, LLM APIs (Gemini, OpenAI, Anthropic), prompt engineering, LLM-as-judge evaluation, embeddings and vector retrieval
Research Methods: Controlled user studies (IRB), large-scale crowdsourcing (MTurk, Prolific), inter-annotator agreement and label aggregation, factorial experimental design, statistical analysis
Systems: FastAPI, React, Django, AWS, Docker, Qdrant, Git, Weights & Biases, L^AT_EX

AWARDS AND HONORS

Best Artifact Award , ACM Conference on AI and Agentic Systems (CAIS)	2026
UC Irvine Dean's Honor List (11 times)	2016 - 2020